

ROBUST LIGHT FIELD SYNTHESIS FROM STEREO IMAGES WITH LEFT-RIGHT GEOMETRIC CONSISTENCY

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ABSTRACT

We propose a lightweight yet effective deep learning pipeline for light field synthesis from a single stereo image pair. Our pipeline consists of a convolutional network (CNN) that enforces a left-right consistency constraint on the light fields synthesized from left and right stereo views, a stage that merges light fields synthesized from left and right stereo views with a novel alpha blending technique, and a final refinement network using a unique 3D convolution operation. Our experiments quantitatively and qualitatively confirm the effectiveness and robustness of the proposed model, which performs favorably against state-of-the-art algorithms for light field synthesis from extremely sparse (only one, two, or four) views while using much fewer parameters.

Index Terms— light field synthesis, light field reconstruction, stereo vision, CNN, deep learning

1. INTRODUCTION

Light field imaging allows users to perform digital refocusing or change the view angle of a scene, and are often acquired through plenoptic cameras [1] or camera arrays [2]. However, capturing high spatial and angular resolution light fields is a great challenge, since plenoptic cameras have inherent spatial-angular resolution tradeoff [1] and since dense camera arrays are impractical for end-users due to bulkiness and cost.

Many techniques have been developed to address the problem of light field reconstruction from sparsely sampled views, or so-called “angular super-resolution” [3]–[5]. These models aim to increase the angular resolution of the sparsely sampled light fields, e.g. reconstruct a 7×7 light field by interpolating 3×3 sampled views. However, the underlying assumption that users are able to acquire an array of sampled views is not always true since such light field capturing devices are not easily accessible.

At the other end of the spectrum, Srinivasan *et al* [6] investigated the problem of light field synthesis from a single RGB image. Although plausible visual results were obtained, reconstructing the absolute depth of a scene from a single image is an ill-posed problem [7]–[9]. Therefore, it would be hard to integrate the synthesis pipeline into applications (e.g. AR) that require absolute depth information. This limits the generalizability of the approach.

In this work, we consider the problem of light field synthesis from stereo image pairs. This is analogous to synthesizing a high-resolution light field from only two horizontal light field views, which is much harder than light field reconstruction from a grid of sampled views. The solution to this problem will be valuable in the real world because, unlike the grid of sampled views, it is relatively easy to capture stereo image pairs either through stereo cameras or by translating a single-lens camera slightly in the horizontal direction. Furthermore, it is convenient to obtain absolute depth from stereo images [10]–[12] for various devices of 3D visual applications, such as AR headsets.

To our knowledge, few works have addressed this problem. Zhang *et al* [13] proposed to use a phased-based synthesis technique to iteratively refine disparity maps between two input stereo views. Then the densely-spaced light field light field views are reconstructed by warping stereo images using the disparity map. However, this iterative approach is inherently slow. Also, it only uses the left view to generate light fields and does not exploit left-right information when synthesizing light fields. Zhou *et al* [14] developed the multiplane-images representation (MPI) technique and synthesized new views by performing homography and alpha composition on the multiplane images. However, this approach requires camera poses of the given stereo images and the views to be synthesized during both training and inference, which may not be available for light fields and stereo images in the wild.

We propose an end-to-end deep learning framework to synthesize light field from stereo images. By exploiting geometric properties of left-right stereo images, our model learns to extrapolate and interpolate stereo images into coarse light fields. It further handles occlusion boundaries and non-Lambertian effects with a novel 3DCNN architecture to refine the results.

The contributions of our work are highlighted as follows:

- Our pipeline synthesizes high quality light field, which is made possible by exploiting left-right consistency of stereo images and distance-weighted alpha blending.
- Our pipeline incorporates a novel patch-wise spatial angular-shuffling 3DCNN network to refine light field using much fewer parameters than prior methods.
- Our pipeline generalizes well for real-world stereo images, and the synthesized light fields lead to promising refocused images.

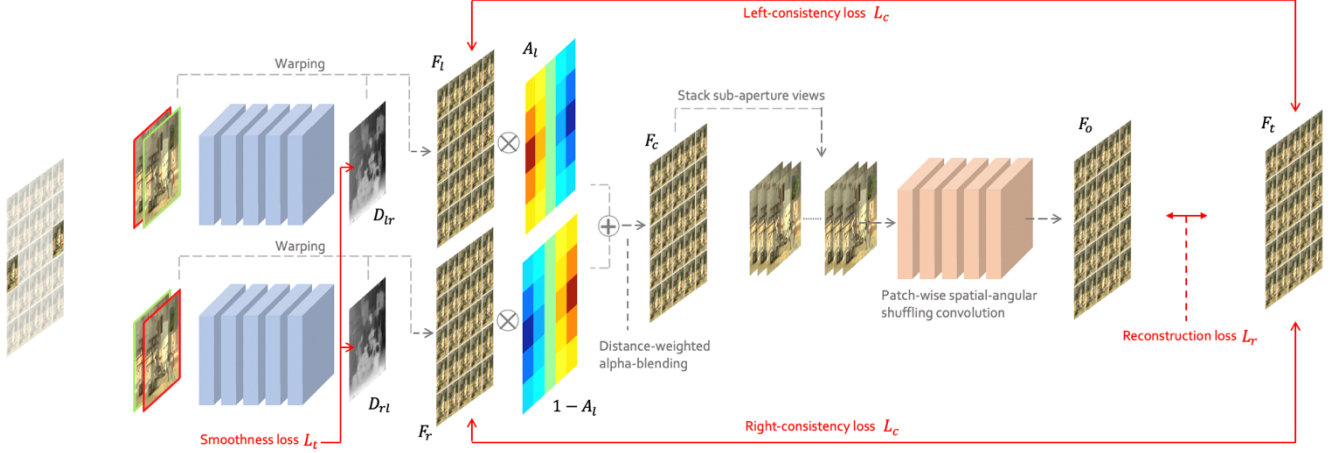


Figure 1. Our light field synthesis pipeline for stereo images consists of three major stages: disparity estimation and warping, light field merging, and light field refinement. The output is a full light field with angular resolution $N \times N$ and spatial resolution $H \times W$.

2. PROPOSED METHOD

2.1. Problem Definition and Notations

We denote the input left stereo view by S_l and the right stereo view by S_r . For each RGB light field view $V(i, j) \in R^{H \times W \times 3}$ in a light field $F \in R^{N \times N \times H \times W \times 3}$ of angular resolution $N \times N$ and spatial resolution $H \times W$, we define $(i, j) \in R^2, 0 \leq i, j < N$ as its *angular coordinates*. We also use S_l and S_r to represent $V(N/2, 0)$ and $V(N/2, N - 1)$, respectively.

2.2. Light Field Synthesis Pipeline

Disparity Estimation and Consistency Constraint

The disparity estimation network, denoted by g , estimates the disparity between two horizontal views. We concatenate the color channels of $V(N/2, 0)$ and $V(N/2, N - 1)$ into a feature map $M_{lr} \in R^{H \times W \times 6}$. The disparity estimation network g takes M_{lr} as input and outputs a horizontal left-to-right disparity map; that is, $g(M_{lr}) = D_{lr} \in R^{H \times W \times 1}$. We then use D_{lr} to backward warp $V(N/2, 0)$ to each light field view $V(i, j)$ and generate the light field F_l .

The light field synthesized from the left view should be the same as the light field synthesized from the right view. To achieve this left-right consistency, we concatenate $V(N/2, N - 1)$ and $V(N/2, 0)$ into a feature map M_{rl} and use the right-to-left disparity map $g(M_{rl}) = D_{rl}$ to warp $V(N/2, N - 1)$ to each light field view $V(i, j)$ to generate the light field F_r . The left-right consistency is imposed in this process through a loss function L_{lr} so that F_l and F_r both approximate the target light field F_t ,

$$L_c = \|F_l - F_t\|_1 + \|F_r - F_t\|_1. \quad (1)$$

Furthermore, we employ a total variation loss widely used for stereo matching and optical flow [15], [16] to smooth the estimated disparity,

$$L_t = \|\nabla_x D_{lr}\|_1 + \|\nabla_y D_{lr}\|_1 + \|\nabla_x D_{rl}\|_1 + \|\nabla_y D_{rl}\|_1. \quad (2)$$

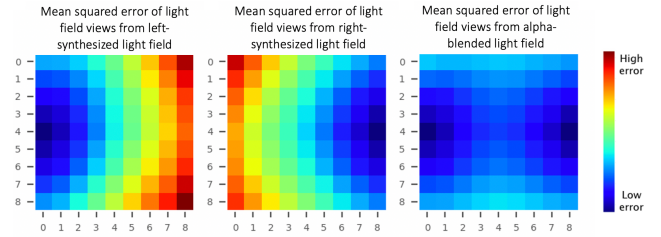


Figure 2. The error between the ground truth view and synthesized view increases as the distance between the synthesized view and input left-right views increases. We mitigate this distance-dependent error by merging F_l and F_r with a novel alpha-blending technique (see the right figure).

Distance-weighted Alpha Blending

To effectively merge the predicted light fields F_l and F_r warped from left and right light field views, we propose a novel *distance-weighted alpha blending* operation. First, we calculate the left distance map $K_l \in R^{N \times N}$ and the right distance map $K_r \in R^{N \times N}$ whose elements are the Euclidean distances between the angular coordinates of each light field view and the input left or right view, that is,

$$K_l(i, j) = \|(i, j) - (N/2, 0)\|_2, \quad (3)$$

$$K_r(i, j) = \|(i, j) - (N/2, N - 1)\|_2. \quad (4)$$

Second, we obtain the *left alpha blending map* $A_l \in R^{N \times N}$, which defines how much the light field views in F_l contribute to the merged light field. A_l is simply acquired by taking the element-wise harmonic mean of K_l and K_r , that is $A_l = \frac{K_r}{K_r + K_l}$. Finally, the merged coarse light field is represented by

$$F_c = F_l \cdot A_l + F_r \cdot (1 - A_l). \quad (5)$$

Light Field Refinement

Table 1. Performance comparison using INRIA, HCI, and Stanford light field datasets. For the comparison with Zhang *et al*, we only consider the synthesized horizontal views in the same row of a $N \times N$ light field as the input stereo views. The bold numbers indicate the best results.

Method	INRIA		HCI		Stanford	
	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
Srinivasan <i>et al</i> (Full light field)	27.43	0.7959	23.60	0.5805	17.75	0.5846
Kalantari <i>et al</i> (Full light field)	27.08	0.7565	23.18	0.5032	23.78	0.7167
Ours (Full light field)	32.18	0.8672	29.73	0.8277	25.85	0.8337
Zhang <i>et al</i> (Horizontal views)	28.87	0.9119	23.06	0.8362	25.72	0.9328
Ours (Horizontal views)	34.18	0.9400	31.50	0.9118	29.06	0.9195

In the final stage of our pipeline, F_c is refined with a refinement network r to handle occlusions and non-Lambertian surfaces. The network r is a novel 3DCNN architecture, which we call Patch-wise Spatial-Angular Shuffling (PSAS) 3DCNN, and consists of five PSAS-convolution blocks.

A PSAS convolution block takes a feature map $M \in R^{N^2 \times H \times W \times 3}$ as input and performs three operations: a 3D convolution, an axis transpose, and another 3D convolution. The first 3D convolution is applied to M to generate a feature map $M' \in R^{N'^2 \times H \times W \times 3}$. Then, M' is reshaped into an $(N' \times N') \times (N' \times H/N') \times (N' \times W/N') \times 3$ tensor with 0th and 2nd axes swapped with 1st and 4th axes and reshaped back to an $N'^2 \times H \times W \times 3$ tensor. Finally, another 3D convolution with $3 \times 3 \times 3$ filters is applied to M' .

We denote the refined light field by $r(F_c) = F_o$ and minimize the reconstruction loss

$$L_r = \|F_o - F_t\|_1 \quad (6)$$

between F_o and the target light field F_t ,

The full objective L sums up the losses in equations (1), (2), and (6). Our learning model optimizes the following objective function:

$$L = w_c \cdot L_c + w_t \cdot L_t + w_r \cdot L_r \quad (7)$$

where w controls the weighting of each loss component.

3. EXPERIEMENTS AND RESULTS

3.1. Datasets

We evaluate our method on three publicly available datasets: INRIA Lytro 1G Dataset [17], HCI 4D Light Field Dataset [18], and the Stanford Light Field Archive [19]. We split INRIA, HCI and Stanford datasets into training/testing data and obtain 46/17, 20/4, 9/2 light fields, respectively. The above datasets have varying maximum disparity between adjacent light field views (INRIA dataset has small disparity while Stanford dataset has large disparity) and can thus help verify if our pipeline is robust to different disparity levels between adjacent light field views.

3.2. Implementation Details

For the disparity estimation network, we employ an architecture similar to the design by Srinivasan *et al* [6]. The final layer of all networks is a $\tanh$ function that scales

output values to $(-1, 1)$ to stabilize the training process. We use ELU [20] as our hidden layer activation function. All networks are trained end-to-end from scratch using the Adam optimization [21] algorithm with default parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^8$. All models use the same loss function weights $w_c = 1$, $w_t = 0.001$, $w_r = 1$. All networks are trained for 2000 epochs with a learning rate of 0.001. We randomly extract 135×135 patches from light field views as training data.

3.3. Quantitative Results

Synthesized Light Field Quality

We evaluate the effectiveness of our model by comparing it with three state-of-the-art light field synthesis methods that work for extremely sparse input views: Zhang *et al* [13] (from two views, non-learning based), Srinivasan *et al* [6] (from center view), and Kalantari *et al* [22] (from four corner views). Table 1 shows the PSNR and SSIM of the synthesized light fields. We can see that the proposed method outperforms the three state-of-the-art light field synthesis methods for all three datasets.

Model Size and Inference Time

We also compare the model size and the inference time to generate a 7×7 angular resolution light field of models trained on the INRIA dataset. As shown in Table 2, our model has 44% fewer parameters compared to Srinivasan *et al*, and is on par with Kalantari *et al*. Since our model synthesizes a whole light field in one pass, its inference time is much shorter than Kalantari *et al*, which synthesizes one view at a time. However, our model has longer inference time than Srinivasan *et al*, since we first synthesize light fields F_l and F_r from left and right light field views, respectively, and then merge them.

Table 2. Inference time and model size comparisons of models trained on the INRIA dataset.

	Model Parameters	Inference time
Srinivasan <i>et al</i>	2.94M	0.113s (GPU)
Kalantari <i>et al</i>	1.64M	34.51s (GPU)
Zhang <i>et al</i>	X (non-learning based)	514.84s (CPU)
Ours	1.74M	0.235s

3.4. Qualitative Results

We also perform visual quality comparisons with Srinivasan *et al* and Kalantari *et al* in Figure 3.

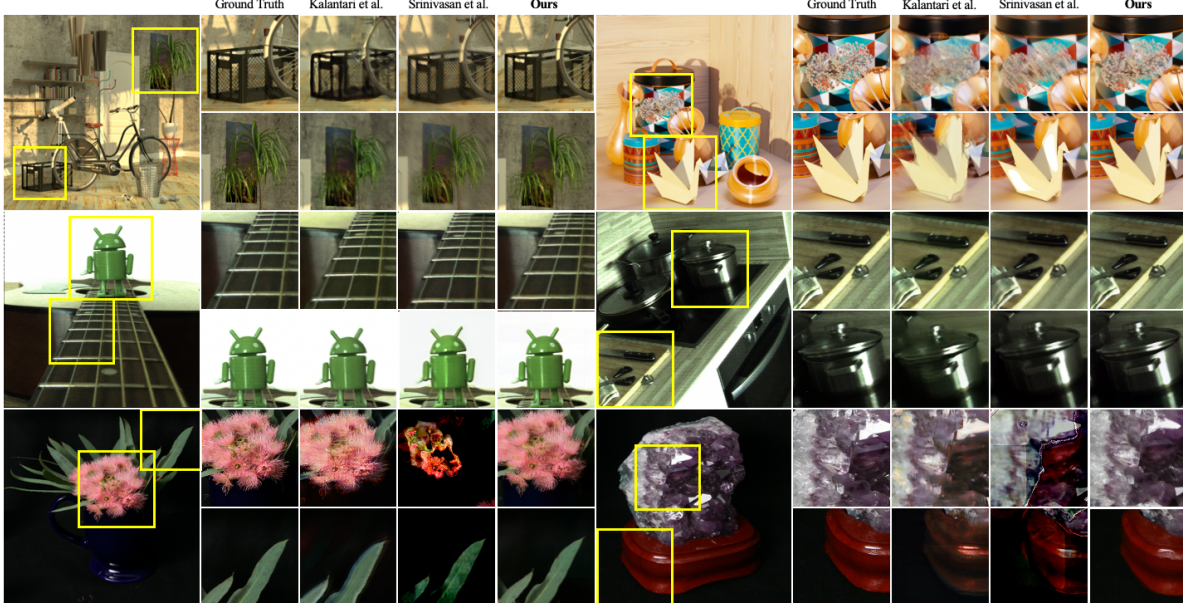


Figure 3. Examples of the synthesized light field views (top-left and right-bottom view from a $N \times N$ light field) from INRIA, HCI, and Stanford light field datasets (listed from top to bottom). Methods of Srinivasan *et al* and Kalantari *et al* are considered for visual quality comparisons. We do not include results of Zhang *et al* here since the code provided by the authors can only synthesize horizontal views.

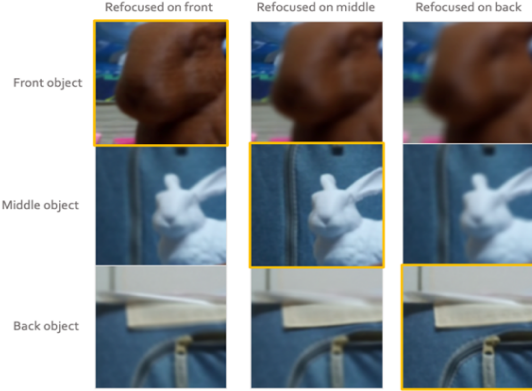


Figure 4. Refocused images of a light field synthesized from stereo images. Each row shows the images of an object as the refocused image plane is placed in the front, middle and back of a scene. Diagonal image patches are in focus.

Generalizability to Real-world Stereo Images

Our model is trained on the INRIA dataset. To verify its generalizability to real-world images, we capture new stereo images by translating a smartphone camera slightly in the horizontal direction. Results in Figure 4 show that the resulting refocused images are visually appealing.

3.5. Ablation Studies

Finally, we conduct ablation studies on models trained on INRIA [17] and HCI [18] datasets to verify the effectiveness of each component in our light field synthesis pipeline. We consider our pipeline excluding left-right consistency and disparity smoothness constraint. We also consider our pipeline using different light field blending methods and refinement network architectures. For the ablation study on

Table 3. Ablation studies of our model trained on INRIA and HCI datasets. The bold numbers denote the best performance.

Method	Metrics (PSNR/SSIM)	
	INRIA	HCI
Ours* (w/o left-right consistency)	32.07/0.8641	29.71/ 0.8338
Ours* (w/o disparity smoothing)	32.11/0.8648	29.48/0.8176
Ours* (F_l synthesized from left view)	30.67/0.8613	27.50/0.8316
Ours* (F_r synthesized from right view)	31.14/0.8307	27.41/0.7105
Ours* ($(F_l + F_r)/2$ average blending)	31.92/0.8659	28.95/0.8019
Ours* (baseline 3DCNN)	31.33/0.8603	29.54/0.8211
Ours* (full model)	32.18/0.8672	29.73/0.8277

network architecture, we compare our refinement network model with the 3DCNN architecture of Srinivasan *et al*. From the results in Table 3, we can see that each component in our light field synthesis pipeline is essential, which means our design is effective.

4. CONCLUSION

We present a lightweight yet effective pipeline to synthesize a light field from a single stereo image pair. The novelty of our method lies in the unique design that fully leverages information from left and right stereo images and a special 3D convolution operation. From only two horizontal views, our pipeline is able to synthesize a 7×7 or 9×9 angular resolution light field. Our method qualitatively and quantitatively outperforms state-of-the-art light field synthesis methods that work for extremely sparse views (one, two or four views). Furthermore, experiments conducted on three light field datasets verify that our pipeline is robust to disparity levels between adjacent light field views. Finally, our pipeline generalizes well to real-world stereo images and can synthesize visually appealing light fields from stereo images in the wild.

5. REFERENCES

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